

# Master Curriculum and Operations Guide

Two camps, 24 primary missions and 8 backups, built on one engineering-design loop: predict, build or solve, test, score, redesign, defend.

## EXECUTIVE SUMMARY

The program runs as two distinct STEM seasons for middle school students: From Trees to Tech with 16 students in 4 teams of 4, and PY-STEM with up to 32 students. From Trees to Tech treats nature as the engineer: trees, leaves, soil, water, ecosystems, sensors, and climate resilience. PY-STEM is broad applied STEM: biomedical engineering, waves, optics, mechanics, materials, accessibility design, and algorithms, with light tie-ins to the Science History Institute and the Charles Library BookBot.

Every primary activity has a measurable win condition and a 100-point rubric. Competition is built into the science task itself, not added on top. The two camps share no core mechanic, build type, or scoring loop.

## FROM TREES TO TECH: 12 PRIMARY ACTIVITIES

| ID     | ACTIVITY                                      | CORE IDEA AND WIN                                                                                                                |
|--------|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| TTT-01 | MudWatt Bioelectric League (Biology + Energy) | Highest combined score from peak reading, daily data quality, controlled-variable design, redesign notes, and final explanation. |
| TTT-02 | Forest Sensor Sprint (Field Science)          | Best evidence-based sensor recommendation with accurate measurements and efficient route planning.                               |
| TTT-03 | Seed Dispersal Derby (Biomimicry)             | Best combined distance, hang time, target landing, biomimicry explanation, and redesign improvement.                             |
| TTT-04 | Xylem Pipeline Relay (Biomimicry)             | Most water delivered with speed, low spill, and strong explanation.                                                              |
| TTT-05 | Greenhouse Climate Controller (Ecology)       | Best plant-zone-control match supported by tour evidence.                                                                        |
| TTT-06 | Pinecone Weather Machine (Biomimicry)         | Most accurate, readable, and fast response after spray-and-dry trials.                                                           |
| TTT-07 | Pollinator Network Draft and Build (Ecology)  | Highest pollinator support across spring, summer, and fall using native and clumping logic.                                      |
| TTT-08 | Arboretum Eco-Quest (Field Science)           | Most accurate checkpoint solutions with safe field conduct.                                                                      |
| TTT-09 | Minecraft Tree World Resilience Cup (Ecology) | Best climate-resilient design with accurate nature-based strategies.                                                             |

| ID     | ACTIVITY                                          | CORE IDEA AND WIN                                                     |
|--------|---------------------------------------------------|-----------------------------------------------------------------------|
| TTT-10 | Tree Ring Climate Detective (Data + Climate)      | Most accurate climate-event inferences with claim-evidence-reasoning. |
| TTT-11 | Lotus Leaf Surface Sprint (Materials)             | Best clean runoff and least residue after one redesign.               |
| TTT-12 | Leaf Stomata Microscope Detective (Field Science) | Most accurate detective ranking with clean data and strong evidence.  |

## PY-STEM: 12 PRIMARY ACTIVITIES

| ID     | ACTIVITY                                     | CORE IDEA AND WIN                                                               |
|--------|----------------------------------------------|---------------------------------------------------------------------------------|
| PYS-01 | Magnetic Capsule Maze Cup (Biomedical)       | Fastest clean run with low penalties and a clear magnetic-control explanation.  |
| PYS-02 | Oobleck Armor Arena (Materials)              | Best target protection with least material and cleanest reset.                  |
| PYS-03 | Cardboard Automata Arcade (Mechanics)        | Most reliable motion with useful mechanism and clear explanation.               |
| PYS-04 | Stethoscope Sprint and Recovery (Biomedical) | Best sound quality plus accurate, respectful data collection.                   |
| PYS-05 | Reaction Time Combine (Biomedical)           | Best median plus most improvement using clean data.                             |
| PYS-06 | SONAR Slinky Showdown (Waves + Sound)        | Most correct wave predictions and station solutions.                            |
| PYS-07 | Pinhole Precision Challenge (Optics + Light) | Best image clarity and concept explanation after one aperture redesign.         |
| PYS-08 | Low-Ropes Force Map Relay (Mechanics)        | Most accurate predictions and clear debrief connections.                        |
| PYS-09 | Hovercraft Hockey Hackathon (Mechanics)      | Best glide plus control in a gym-safe tournament.                               |
| PYS-10 | Spectra Sleuth Showdown (Optics + Light)     | Most correct spectrum matches and strongest safe fireworks-science explanation. |
| PYS-11 | BookBot Bin Logic Challenge (Computing)      | Most correct retrievals with few traffic conflicts and clear algorithm defense. |
| PYS-12 | Accessibility Ramp Rescue Lab (Design)       | Best safe ramp meeting slope, load, portability, and client criteria.           |

BACKUP ACTIVITIES

Held ready for weather, trip delays, or materials failure. Same loop, lighter setup.

| ID     | ACTIVITY                                           | CORE IDEA AND WIN                                  |
|--------|----------------------------------------------------|----------------------------------------------------|
| TTB-01 | Root Grip Erosion Rescue (Field Science)           | Most soil retained after standard pour.            |
| TTB-02 | Tree Height Triangulation Shootout (Field Science) | Closest to instructor reference height.            |
| TTB-03 | Urban Heat Shade Dash (Data + Climate)             | Best data-backed cool route.                       |
| TTB-04 | Photosynthesis Float-Off Playoffs (Field Science)  | Fastest half-float time with controlled variables. |
| PYB-01 | Pencil Pressure Safety Lab (Mechanics)             | Largest area protected with clear data.            |
| PYB-02 | Domino Neuron Relay (Biomedical)                   | Fastest reliable signal with best explanation.     |
| PYB-03 | Pulley Rescue Relay (Mechanics)                    | Best lift with correct setup and explanation.      |
| PYB-04 | Barcode Checksum Rescue (Computing)                | Most bad codes detected with few false alarms.     |

## THE SHARED LEARNING LOOP

---

Every activity moves teams through the same cycle, which mirrors the engineering design process and the practices in NGSS MS-ETS1.

- Predict: teams commit to a prediction and a reason before any materials move.
- Build or solve: teams construct or work the challenge using approved materials.
- Test: teams run a standard test and record a baseline result.
- Score: the 100-point rubric rewards design, data quality, teamwork, and explanation.
- Redesign: teams change one variable, predict its effect, and re-test.
- Defend: teams justify their final claim with specific evidence.

## SAFETY OVERVIEW

---

No student open flames, dry ice, high voltage, hazardous chemicals, pressure vessels, dangerous projectiles, major staining, or cut-open glow sticks are used anywhere in the program.

- All electrical work is at battery voltage. MudWatt cells are built over a tray and handled with gloves.
- Magnets are counted and controlled. Oobleck and water activities use trays and immediate cleanup.
- Pinhole optics include a no-direct-sun-view rule. Field trips follow host rules first.
- PPE is required where splash, soil, oobleck, plant peel, or cleanup risk exists: goggles, nitrile gloves, handwashing.
- Allergy notes appear in each activity guide. The full allergy roster is confirmed before camp.

Detailed daily tasks live in the Staff Setup, Prep, and Safety Checklist.

## OPERATIONS AND ROLES

---

- Core rule: bins packed before instruction starts, score sheets ready before testing starts, PPE visible before materials are distributed.
- Each team rotates roles: captain, safety lead, builder, tester, data lead, presenter.
- Best 9 of 12 primary scores count toward the final leaderboard so a single rough activity does not sink a team.
- Backups are staged and ready every day, not improvised on the spot.

## MATERIALS AND PACKING BINS

---

Each activity has a labeled bin, with reusable and consumable items tracked per activity.

## PRODUCTION NOTES

---

- Leaf chromatography and tie-dye were removed; Leaf Stomata Microscope Detective is used instead.
- BookBot is framed as automated storage and retrieval, bin addressing, and routing logic, not conveyor throughput.
- PY-STEM stays broad, with only selected anchor tie-ins to the Science History Institute and the BookBot.
- The two camps were audited against each other and against the legacy no-repeat list; no banned activity is reused.